



VARUNA OPS

DRONE OPERATIONS PLATFORM

CLOUD DEPLOYMENT PROPOSAL

Production Deployment Roadmap & Pricing Plan

A complete, phased plan to take the Varuna Ops platform live on the cloud — covering where the frontend, backend, database, queue and file storage are hosted, the exact tools used at each layer, capacity & storage estimates, and transparent monthly pricing across three budget tiers.

 React + Vite Frontend

 Node.js API + Worker

 PostgreSQL

 Redis Queue

 Cloudflare R2 Storage

PREPARED FOR

Client / Stakeholder Review

DOCUMENT

Deployment & Pricing v1.0

DATE

June 2026

Executive Summary & System Architecture

Varuna Ops is a **file-heavy, real-time drone operations platform**. It is not a single app — it is a set of cooperating services that must each be hosted on the right kind of infrastructure. The most important architectural fact for budgeting: each drone project produces **50–100 GB of raw imagery & deliverables**, so **object storage** (not the server or database) is the component that grows the cost over time. We host that on **Cloudflare R2**, which charges for storage but — critically — has **zero egress fees**, so clients downloading large deliverables never trigger surprise bandwidth bills.

RECOMMENDED MONTHLY

\$85–95/mo

≈ ₹7,300–8,200 · managed, low-ops

LEANEST START

\$11–14/mo

≈ ₹950–1,200 · single server

GO-LIVE TIMELINE

5–6 days

From zero to production

Deployment Topology — what runs where

LAYER 1 · EDGE

React + Vite Frontend

Static build (HTML/JS/CSS). Served from a global CDN — fast everywhere.

USERS

Browser (Admin / PM / Pilot)

Web app + real-time Socket.IO updates & notifications.

▼ HTTPS / WebSocket (API calls) ▼

LAYER 2 · COMPUTE

API Service (Express)

HTTP API, auth, Socket.IO. Process: *varuna-api*.

LAYER 2 · COMPUTE

Background Worker

ZIP/export jobs, emails, scheduler. Process: *varuna-worker*.

▼ data · jobs · files ▼

LAYER 3 · STATE

PostgreSQL

Projects, users, pipeline, estimations, notifications.

LAYER 3 · STATE

Redis

BullMQ job queue + scheduler lease.

LAYER 3 · STATE

Cloudflare R2

Raw drone data & deliverables (the big one — 50–100 GB/project).

▼ outbound ▼

LAYER 4 · SERVICES

Email (SMTP / Resend)

Welcome, allocation, approval & expiry notifications.

LAYER 4 · SERVICES

Cloudflare DNS + SSL

Domain, free TLS certificate, CDN, WAF.

Why two compute processes? Your `ecosystem.config.js` already splits the app into an **API** and a **worker**. Heavy jobs (zipping a 100 GB project for download, exports, scheduled license/maintenance checks) run on the worker so they never slow down or crash the live API. Every hosting tier below keeps this separation.

Every Tool & Where Each Piece Deploys

This is the complete list of platforms involved. For each layer we list the **recommended choice** plus solid alternatives, so the decision is transparent.

Layer	What deploys here	Recommended tool	Good alternatives	Role
Frontend	React + Vite static build	CLOUDFLARE PAGES	Vercel · Netlify	Global CDN hosting of the web app
Backend API	Express + Socket.IO (<i>varuna-api</i>)	RENDER / Railway	Fly.io · Hetzner VPS · AWS ECS	Serves the HTTP API & live updates
Worker	BullMQ jobs + scheduler (<i>varuna-worker</i>)	RENDER WORKER	Railway · Fly.io · same VPS	ZIP/export, emails, cron checks
Database	PostgreSQL	NEON / Supabase	Render PG · DO Managed · AWS RDS	Primary relational data store
Queue	Redis	UPSTASH REDIS	Render Key-Value · self-hosted	Background job queue
File storage	Drone data & deliverables	CLOUDFLARE R2 (IN USE)	Backblaze B2 · AWS S3	Large object storage, free egress
Email	Transactional notifications	RESEND / Brevo	AWS SES · SendGrid · Gmail SMTP	Outbound system emails
DNS / SSL / CDN	Domain & certificates	CLOUDFLARE	Namecheap + Let's Encrypt	Routing, free TLS, DDoS shield
CI / CD	Auto-deploy on git push	GITHUB + ACTIONS	Platform-native git deploys	Build & release pipeline
Monitoring	Uptime & error tracking	UPTIMEROBOT + SENTRY	Better Stack · platform logs	Alerts when something breaks
Process mgr (VPS only)	Keeps API+worker alive	PM2 (CONFIGURED)	Docker Compose · systemd	Restart on crash, log rotation

Already decided for you: The codebase already uses **Cloudflare R2** for storage and ships a **PM2** process file. So storage and process-management tooling are settled — we only choose the hosting tier.

🔑 One-time setup assets

- Domain name (e.g. *varunaops.com*)
- Cloudflare account (DNS + Pages + R2)
- Strong JWT_SECRET (`openssl rand -hex 64`)
- R2 bucket + access keys
- SMTP / Resend API key

🔒 Security must-dos at launch

- Rotate any DB password / secret ever shared
- Never commit the real `.env`
- Restrict CORS to your domain (`FRONTEND_URL`)
- Enable automated database backups
- Force HTTPS everywhere (free via Cloudflare)

Three Deployment Tiers

Pricing scales with ambition. Pick based on expected users and how much you want to manage yourself. **All three keep the same architecture** — they differ only in *where* the pieces run and how much is managed for you.

TIER 1

Starter / Pilot

\$11–14/mo

≈ ₹950–1,200/mo

One VPS runs everything (Hetzner) via Docker: API + worker + Postgres + Redis + Nginx.

- ✓ Cheapest possible
- ✓ Full control
- ✗ You manage backups, updates, security
- ✗ Single point of failure

Best for: demo, internal pilot, ≤ ~30 users.

TIER 2 · RECOMMENDED

Growth / Managed

\$85–95/mo

≈ ₹7,300–8,200/mo

Managed cloud services — each layer on a purpose-built platform with auto-backups & git deploys.

- ✓ Managed backups & scaling
- ✓ Auto-deploy on git push
- ✓ Minimal ops burden
- ✗ Higher per-unit cost than a VPS

Best for: real production launch, paying clients, hundreds of users.

TIER 3

Enterprise / Scale

\$280–380/mo

≈ ₹24,000–32,700/mo

AWS high-availability: multi-AZ database, load-balanced API, auto-scaling worker fleet.

- ✓ High availability & SLA
- ✓ Auto-scales to thousands of users
- ✓ Compliance-friendly
- ✗ Complex — needs a DevOps owner

Best for: large org, strict uptime, heavy concurrent load.

Our recommendation: Launch on **Tier 2 (Growth / Managed)**. It gives you real production reliability — automated backups, zero-downtime deploys, and managed databases — without hiring a DevOps engineer. You can start Tier 1 for a demo and move up, or jump to Tier 3 only once concurrent load genuinely demands it. **Storage (R2) is billed separately on all tiers** and is detailed in Section 5.

Note on scaling the API: Today the API runs as a single instance (correct for Socket.IO without a shared adapter). To run multiple API instances behind a load balancer (Tier 3), the codebase needs the `@socket.io/redis-adapter` wired in — a small, well-understood change already noted in the project's own deployment comments.

Detailed Pricing Breakdown

Tier 1 — Starter / Pilot

SINGLE VPS

Component	Service	USD / mo	INR / mo	Notes
Server (all-in-one)	Hetzner CX32 — 4 vCPU, 8 GB, 80 GB SSD	\$9.00	₹775	API+worker+PG+Redis in Docker
Frontend	Cloudflare Pages	\$0.00	₹0	Free — unlimited bandwidth
Object storage	Cloudflare R2 (≈100 GB)	\$1.50	₹130	Grows with data (Section 5)
Email	Resend / Brevo (free tier)	\$0.00	₹0	Up to 3,000 emails/mo free
DNS / SSL / CDN	Cloudflare (free plan)	\$0.00	₹0	Free TLS + DDoS shield
Domain	.com (amortized)	\$1.00	₹86	≈ \$12/yr
TOTAL — Tier 1		≈ \$11.50	≈ ₹990	+ storage as data grows

Tier 2 — Growth / Managed

RECOMMENDED

Component	Service	USD / mo	INR / mo	Notes
Backend API	Render Web Service — 2 GB RAM	\$25.00	₹2,150	Always-on, auto-deploy
Worker	Render Background Worker — 512 MB	\$7.00	₹600	Bump to \$25 if exports heavy
Database	Neon Postgres — Launch (10 GB)	\$19.00	₹1,635	Auto-backup + branching
Queue	Upstash Redis	\$10.00	₹860	Free tier may suffice early
Frontend	Cloudflare Pages	\$0.00	₹0	Free
Object storage	Cloudflare R2 (≈250 GB)	\$3.75	₹325	Scales with projects
Email	Resend Pro (50k emails)	\$20.00	₹1,720	Brevo \$9 is a cheaper option
DNS / SSL / Domain	Cloudflare + .com	\$1.00	₹86	Free DNS, domain amortized
TOTAL — Tier 2		≈ \$85.75	≈ ₹7,375	+ storage as data grows

Tier 3 — Enterprise / Scale

AWS HA

Component	Service	USD / mo	INR / mo	Notes
Frontend	S3 + CloudFront CDN	\$10.00	₹860	Usage-based
Backend API	ECS Fargate — 2 tasks (HA)	\$90.00	₹7,740	Load-balanced

Component	Service	USD / mo	INR / mo	Notes
Worker	ECS Fargate — 1 task	\$35.00	₹3,010	Auto-scalable
Load balancer	Application Load Balancer	\$20.00	₹1,720	+ sticky sessions for sockets
Database	RDS Postgres — t4g.medium Multi-AZ	\$130.00	₹11,180	HA failover + backups
Queue	ElastiCache Redis — t4g.small	\$30.00	₹2,580	Managed Redis
Object storage	Cloudflare R2 (≈ 1 TB)	\$15.00	₹1,290	Kept on R2 (free egress)
Email	AWS SES	\$5.00	₹430	\$0.10 / 1,000 emails
DNS / WAF / Domain	Route 53 + Cloudflare	\$12.00	₹1,030	Routing + firewall
TOTAL — Tier 3		≈ \$347	≈ ₹29,840	+ storage as data grows

PRICING DISCLAIMER: Figures are indicative list prices as of **early 2026** and exclude taxes/GST. Cloud providers adjust pricing and free-tier limits over time, and actual cost varies with real traffic, compute hours and storage volume. INR shown at ≈ ₹86/USD. Always confirm current pricing on each provider's site before committing. Storage (Cloudflare R2) is the main variable cost and is detailed next.

Storage, Space & Usage — How Much Can It Hold?

Two very different kinds of "space" matter. The **database** stays small (records are tiny); the **object storage** is what grows, because every drone project carries tens of gigabytes of imagery and deliverables. Plan the budget around object storage.

Database (PostgreSQL) — stays small

Holds text records only: projects, users, pipeline, estimations, notifications.

- **10 GB** handles **tens of thousands** of projects & users comfortably
- Typical real usage in year one: **under 1–2 GB**
- Redis queue: a few hundred MB at most

Object storage (R2) — the one that grows

Holds raw drone data & deliverables. This is the real capacity question.

- Each project $\approx$ **50–100 GB** (the platform is built for this)
- **No hard cap** — R2 bills per GB, grows infinitely
- **Egress is FREE** — client downloads cost nothing

Cloudflare R2 storage cost by volume (at \$0.015 / GB-month, free egress)

Projects stored	Approx. data ($\approx$ 75 GB ea.)	Storage \$/mo	Storage ₹/mo	\$/yr	Comment
Up to $\sim$ 13 GB	Free tier	\$0.00	₹0	\$0	First 10 GB free
10 projects	$\approx$ 0.75 TB	\$11.25	₹970	\$135	Small operation
25 projects	$\approx$ 1.9 TB	\$28.10	₹2,420	\$337	Growing
50 projects	$\approx$ 3.75 TB	\$56.25	₹4,840	\$675	Busy team
100 projects	$\approx$ 7.5 TB	\$112.50	₹9,675	\$1,350	Large operation
250 projects	$\approx$ 18.75 TB	\$281.25	₹24,190	\$3,375	Enterprise archive

The free-egress advantage in numbers: On AWS S3, a client downloading a 100 GB project deliverable would cost $\approx$ **\$9 in egress every time**. On **Cloudflare R2 that download is \$0**. For a platform whose whole point is delivering big files to clients, this single fact can save thousands of dollars a year — which is exactly why R2 was chosen.

Concurrent-user capacity (compute)

Tier	Comfortable concurrent users	What limits it	How to scale up
Tier 1 VPS	$\sim$ 30–80 active	Single 8 GB server shared by all services	Bigger VPS, or split DB out
Tier 2 Managed	$\sim$ 200–500 active	Single API instance (Socket.IO)	Add Redis socket-adaptor $\rightarrow$ multi-instance
Tier 3 AWS	Thousands	Effectively horizontal — add tasks	Auto-scaling group, larger RDS

"Active" = simultaneously interacting. Total registered users can be far higher; most aren't online at once. Large file uploads use resumable chunked sessions (up to 72 h) so slow links and 50–100 GB datasets are handled gracefully.

Step-by-Step Deployment Roadmap

A clear, phased path from an empty cloud account to a live, secured production system. Total effort: roughly **5–6 working days** for the recommended (Tier 2) setup.

- Phase 0 — Pre-flight & accounts** ~1 day
Buy the domain. Create Cloudflare, hosting, database & email accounts. Create the R2 bucket + access keys. Generate a strong `JWT_SECRET`. **Rotate any secret ever shared**. Draft the production `.env` (never commit it).
- Phase 1 — Database** ~½ day
Provision managed PostgreSQL. Run migrations (`npm run migrate` — all 16+). Seed the first admin user. Confirm automated backups are on.
- Phase 2 — Backend API + Worker** ~1 day
Deploy *varuna-api* and *varuna-worker* as two services. Set all env vars (DB, R2, Redis, SMTP). Connect Redis & R2. Verify the `/health` endpoint and that a test job runs on the worker.
- Phase 3 — Frontend** ~½ day
Set `VITE_API_URL` to the API domain. Build (`vite build`) and deploy to Cloudflare Pages. Attach the custom domain.
- Phase 4 — Networking, domains & SSL** ~½ day
Point DNS for app + API subdomains. Enable free TLS. Lock CORS & Socket.IO origin to your domain via `FRONTEND_URL`. Confirm real-time updates work end-to-end.
- Phase 5 — Hardening & observability** ~1 day
Set up daily DB backups, uptime monitoring (UptimeRobot), error tracking (Sentry), and email-delivery checks. Verify rate-limiting & security headers (Helmet) are active in production.
- Phase 6 — Go-live & handover** ~½ day
Full smoke test (login, project create, upload, export/ZIP, notifications, emails). Final DNS cutover. Hand over credentials, runbook & monitoring dashboards.

✅ Production-readiness today

The platform has been through multiple zero-bug stabilization passes — crashes fixed, migrations applied, security headers, rate-limiting, JWT auth, 2FA, RBAC and resumable uploads are all in place. Internal readiness is assessed at **~99%**.

📋 Small pre-launch checklist

- Move background jobs to live Redis (currently optional)
- Add Socket.IO Redis adapter before multi-instance
- Cap ZIP memory for very large projects
- Plan Node 22 upgrade (AWS SDK, by 2027)

Recommendation & Total Cost of Ownership

For a real launch with paying clients, deploy on **Tier 2 (Growth / Managed)**. It delivers production-grade reliability — managed databases, automated backups, zero-downtime git deploys — without the overhead of running servers yourself, and it costs a fraction of the enterprise tier.

Tier	Infra \$/mo	Infra ₹/mo	+ Storage (50 proj.)	≈ Total ₹/mo	Verdict
Tier 1 Starter	\$11.50	₹990	+\$56	₹5,800	Demo / pilot
Tier 2 Growth ★	\$85.75	₹7,375	+\$56	₹12,200	Recommended launch
Tier 3 Enterprise	\$347	₹29,840	+\$56	₹34,700	Large-scale / HA

Storage column assumes 50 active projects (~3.75 TB on R2). Storage is shared logic across tiers; it grows with your data, not with the tier.

Why this plan is low-risk

- **Start small, grow later** — begin on Tier 1/2, move up only when load demands
- **No vendor lock on storage** — R2 is S3-compatible; portable to S3/B2
- **Predictable bills** — free egress removes the scariest variable cost
- **Fast to launch** — ~5–6 days, codebase already production-hardened

Recommended stack at a glance

Frontend	Cloudflare Pages
API + Worker	Render
Database	Neon PostgreSQL
Queue	Upstash Redis
Storage	Cloudflare R2
Email	Resend
DNS/SSL	Cloudflare

NEXT STEP

Approve a tier & we begin Phase 0

On approval we provision accounts, deploy all services, secure the domain with SSL, enable backups & monitoring, and hand over a live, documented production environment within roughly one working week.